MACHINE LEARNING LAB

MID PROJECT

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🎬 Movie Recommendation System using Machine Learning

1. Introduction

A **Movie Recommendation System** is an application of Machine Learning that helps users discover movies based on their preferences and past behavior. With the rapid growth of digital platforms such as Netflix and Amazon Prime Video, users are exposed to thousands of choices, making it difficult to select relevant content.

Recommendation systems solve this problem by analyzing user data and suggesting personalized items. These systems are widely used in entertainment, e-commerce, and social media.

2. Problem Statement

The objective of this project is to design a system that predicts movie ratings using **Supervised Learning** techniques.

Instead of directly recommending movies, the system:

- Predicts the rating a user would give
- Uses this prediction to recommend high-rated movies

3. Dataset Description

The dataset used is the **MovieLens dataset** from Kaggle.

Features:

- **userId** → Unique user identifier
- **movieId** → Unique movie identifier
- **rating** → User rating (1–5)
- **timestamp** → Time of rating
- **title** → Movie name
- **genres** → Movie categories

This dataset is widely used in recommendation system research.

4. Data Preprocessing

Before applying machine learning models, the dataset must be cleaned and transformed.

Steps:

1. Remove missing values
2. Extract year from movie titles
3. Convert genres into numerical form
4. Merge datasets (movies + ratings)

This step ensures that the data is suitable for modeling

5. Exploratory Data Analysis (EDA)

EDA is used to understand patterns and relationships in the dataset.

Key Analysis:

- Distribution of ratings
- Most popular movies
- Trends over time

Visualization Techniques:

- Histogram (rating distribution)
- Line graph (rating trends over years)
- Bar charts (model comparison)

EDA helps in making better modeling decisions.

6. Feature

Feature engineering improves model performance by transforming data.

Techniques Used:

- **Genre Encoding** using CountVectorizer
- Conversion of categorical data into numerical form

This allows machine learning algorithms to process the data effectively.

7. Algorithms Used

7.1 Linear Regression

Linear Regression is one of the simplest machine learning algorithms.

Concept:

It assumes a linear relationship between input variables and output.

Advantages:

- Simple and fast
- Easy to interpret

Disadvantages:

- Cannot handle complex patterns
- Lower accuracy on non-linear data

7.2 Random Forest Regressor

Random Forest is an ensemble learning algorithm.

Concept:

- Combines multiple decision trees
- Uses averaging to improve performance

Advantages:

- Handles non-linear relationships
- High accuracy
- Reduces overfitting

Disadvantages:

- Slower than simple models
- More computational cost

8. Evaluation Metrics

To evaluate model performance, we use:

◆ Mean Squared Error (MSE)

- Measures prediction error
- Lower value = better model

◆ R^2 Score

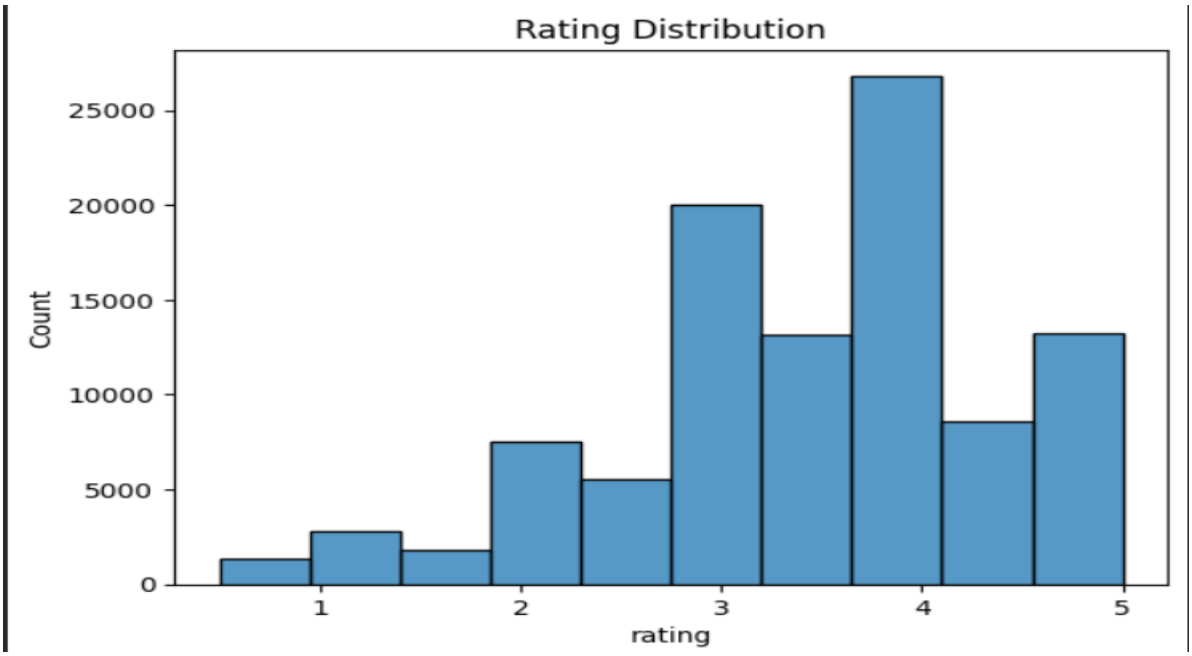
- Measures how well the model explains variance
- Value ranges from 0 to 1
- Higher value = better model

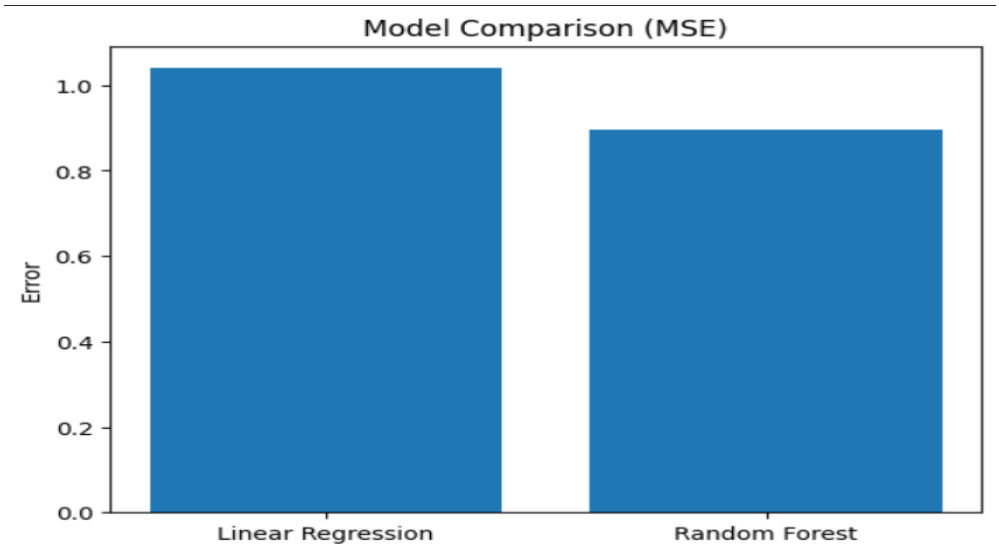
9. Results and Comparison

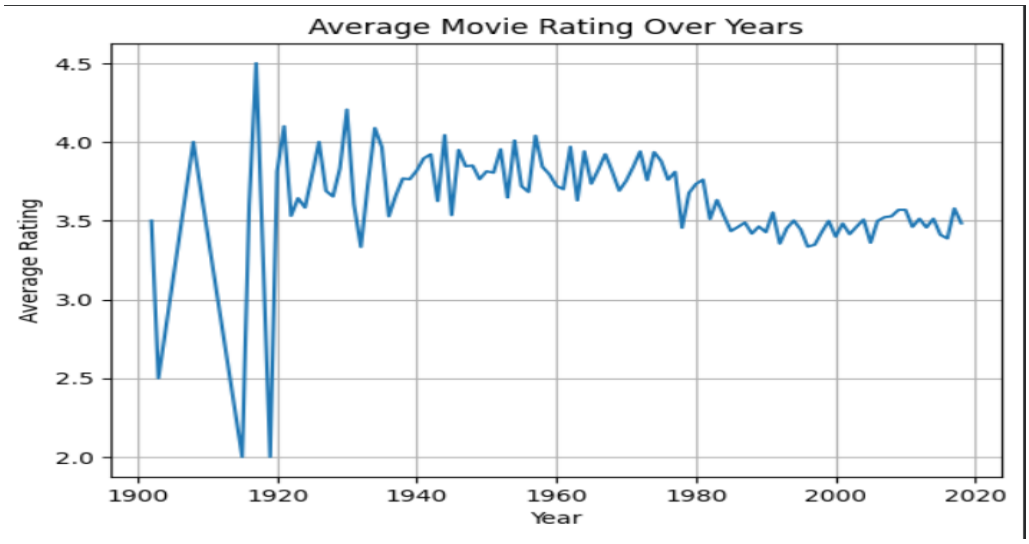
Model	Performance
Linear Regression	Moderate
Random Forest	High

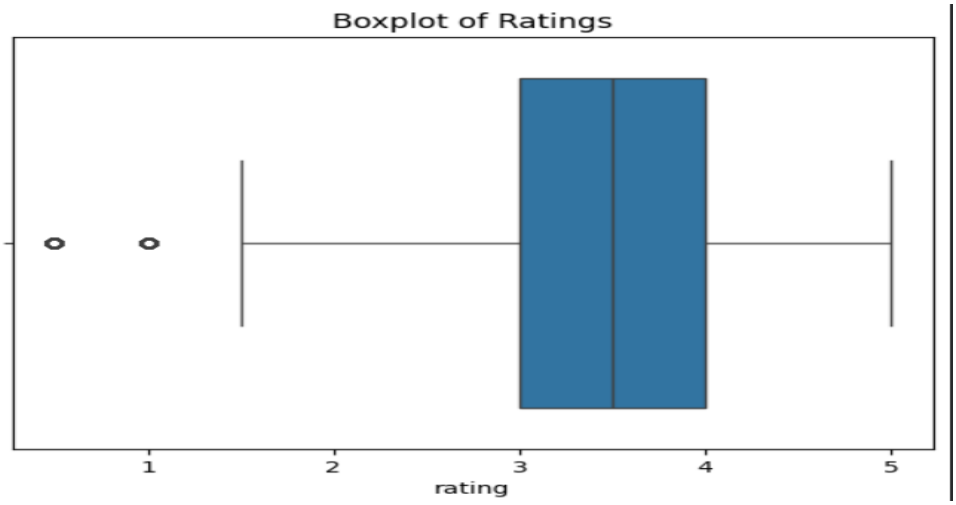
Analysis:

- Linear Regression performs well for simple patterns
- Random Forest captures complex relationships




Model Comparison (MSE)
Error
1.0
0.8
0.6
0.4
0.2
0.0
Linear Regression
Random Forest


Average Movie Rating Over Years
Average Rating
4.5
4.0
3.5
3.0
2.5
2.0
1900
1920
1940
1960
1980
2000
2020
Year


Boxplot of Ratings
1
2
3
4
5
rating

10. Graphical Analysis

Graphs used in the project:

1. Rating Distribution Histogram
2. Model Comparison Bar Chart
3. Average Rating Over Years (Line Graph)

These graphs help in understanding data trends and model performance.

11. Advantages of the System

Personalized recommendations
Handles large datasets
Improves user experience
Scalable for real-world applications

12. Conclusion

This project successfully demonstrates a **Movie Recommendation System using Machine Learning**. Two supervised learning algorithms were implemented and compared.

- Random Forest performed better than Linear Regression
- The system can predict movie ratings effectively
- It can be extended to real-world applications

GITUP LINK: https://github.com/HARIS2003JULY/machinelearnigMIdProject